

Embedded systems I / embedded programming

The timers

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v2.0

Timers : Objectives

Working with time (Timers)

Timers
- **Overview**
- **Configuration**

PWM module
- **Usage and configuration**



Timers : why ?

Often, we need to measure time between events or we want to generate periodic events (to wait)

Until now, either:

- **Active wait (`HAL_Delay()` for example)**
- **Count cycles**

Problems ?

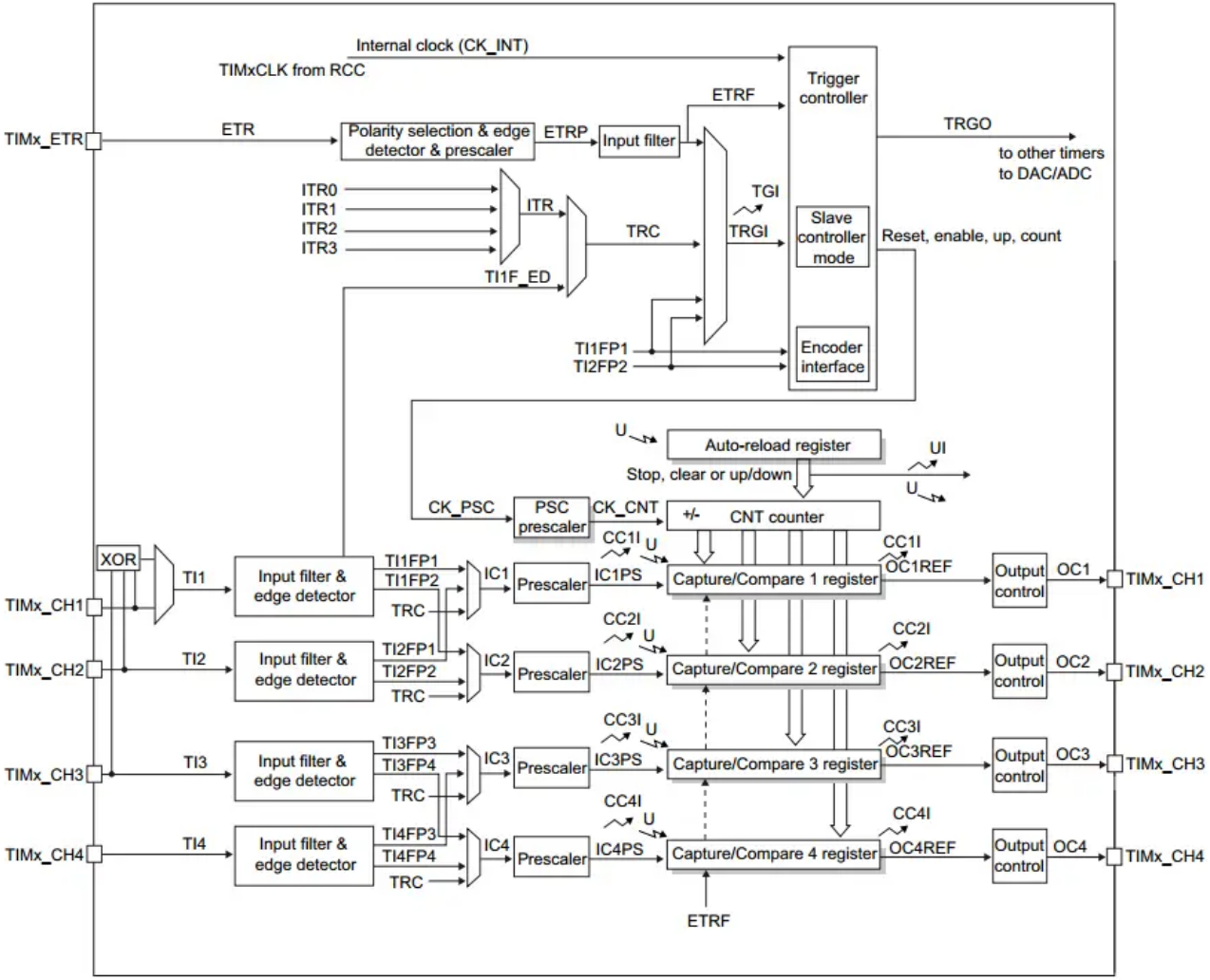
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Timers : what is it ?

Independent peripheral that count time (dedicated hardware)

- 2 advanced-control timers (TIM1/TIM8) – 16 bits
- 7 general-purpose timers (TIM2..TIM5, TIM15..TIM17) – 32 bits
- 2 basic timers (TIM6/TIM7) – 16 bits
- 4 low-power timers (LPTIM1..LPTIM4) – 16 bits
- Clock can be Internal clock / external / divided, …
- Dedicated trigger
- input / output functions (capture/compare/PWM)

Timers : general purpose timer

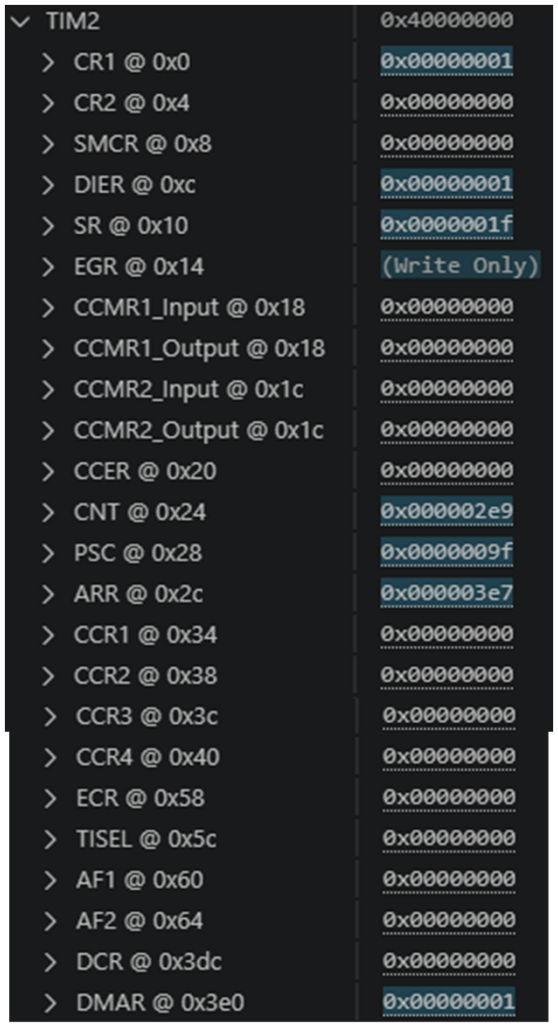


Timers : A lot of registers

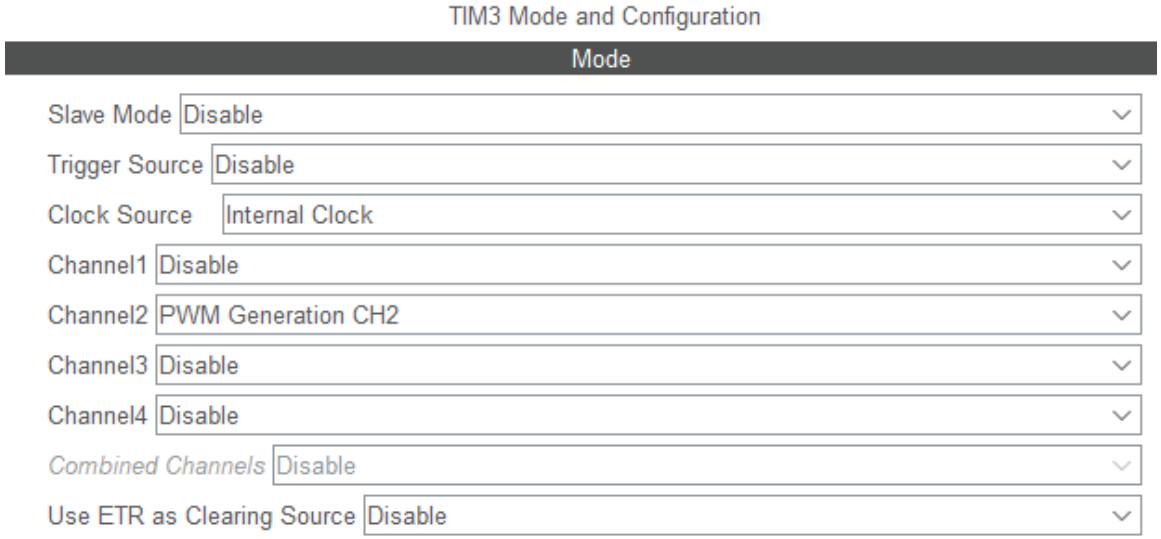
Timer2 (general purpose):
- 24 registers

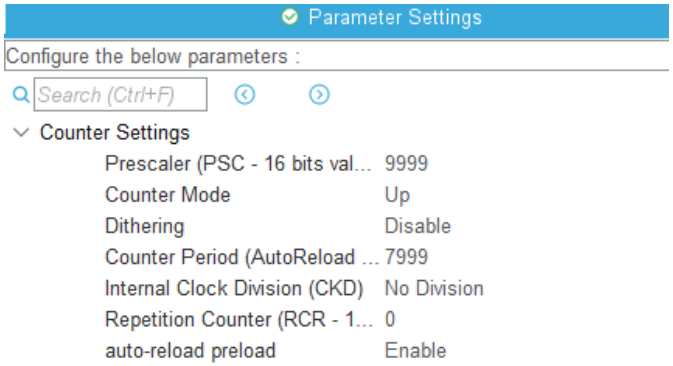
Timer1 (advanced):
- 30 registers

Timer6 (basic):
- 8 registers



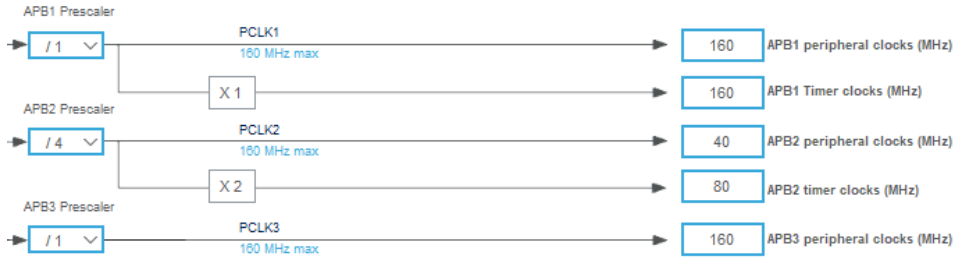
Timers : Basic registers / settings





Timers : APB clock

If APBx prescaler is not 1: You must see datasheet to know the input frequency (is it APB1 or APB2 ?)



31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	USART 6EN	CRSEN	Res.	I2C2EN	I2C1EN	UART5 EN	UART4 EN	USART 3EN	USART 2EN	Res.
						rw	rw		rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SPI2E N	Res.	Res.	WWDG EN	Res.	Res.	Res.	Res.	Res.	TIM7E N	TIM6E N	TIM5E N	TIM4E N	TIM3E N	TIM2E N
	rw			rs						rw	rw	rw	rw	rw	rw

APB1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	DSIEN	LTDCE N	GFXTI MEN	USBEN	Res.	SAI2E N	SAI1E N	Res.	Res.	TIM17E N	TIM16E N	TIM15E N
				rw	rw	rw	rw		rw	rw			rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	USART 1EN	TIM8E N	SPI1E N	TIM1E N	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	rw	rw	rw	rw											

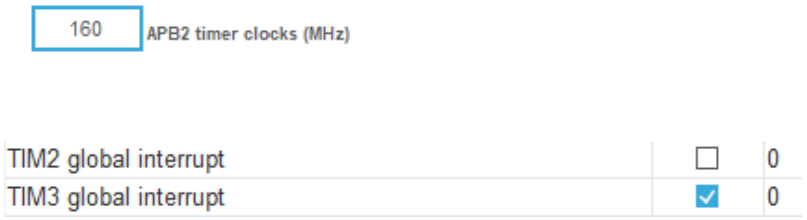
APB2

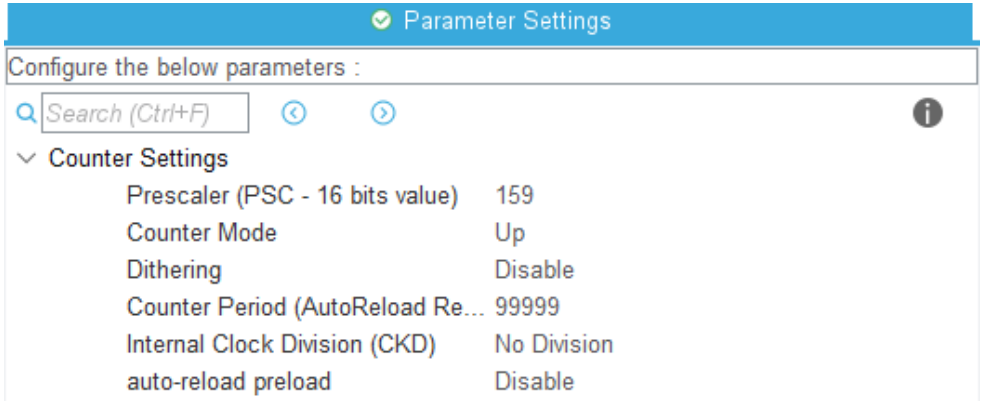
Timers : periodic event

To generate a periodic event, define the period

Clock source is internal

- Period is defined by timers CLK, prescaler, counter period
- Can generate interrupt





Timers : periodic event 1/2

Start the timer

```
...
/*Start the timer in interrupt mode*/
HAL_TIM_Base_Start_IT(&htim3);
...
```

main.c

Implement interrupt (ugly…)

```
...
void TIM3_IRQHandler(void)
{
/* USER CODE BEGIN TIM3_IRQn 0 */
/* USER CODE END TIM3_IRQn 0 */
HAL_TIM_IRQHandler(&htim3);
/* USER CODE BEGIN TIM3_IRQn 1 */
/* USER CODE END TIM3_IRQn 1 */
}
...
```

stm32u5xx_it.c

Timers : periodic event 2/2

Start the timer

```
...
/*Start the timer in interrupt mode*/
HAL_TIM_Base_Start_IT(&htim3);
...
```

main.c

Implement interrupt (very nice)

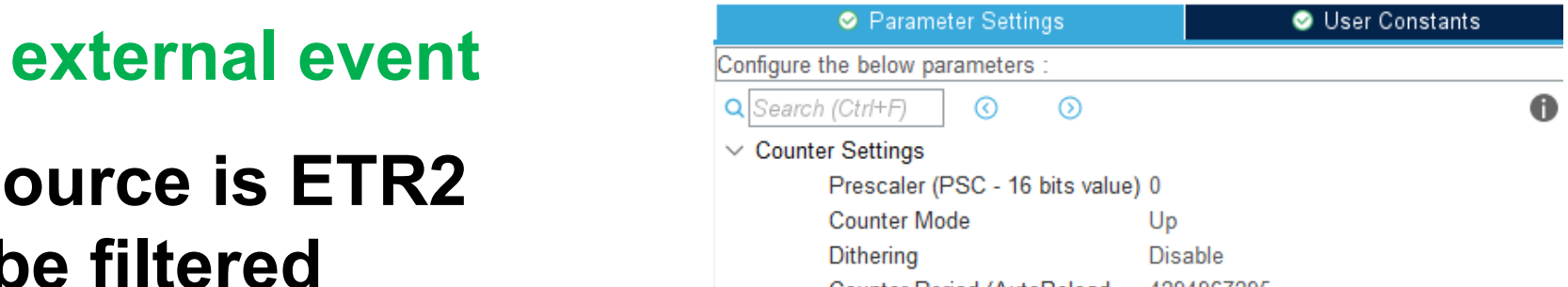
```
...
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
  ...
}
...
```

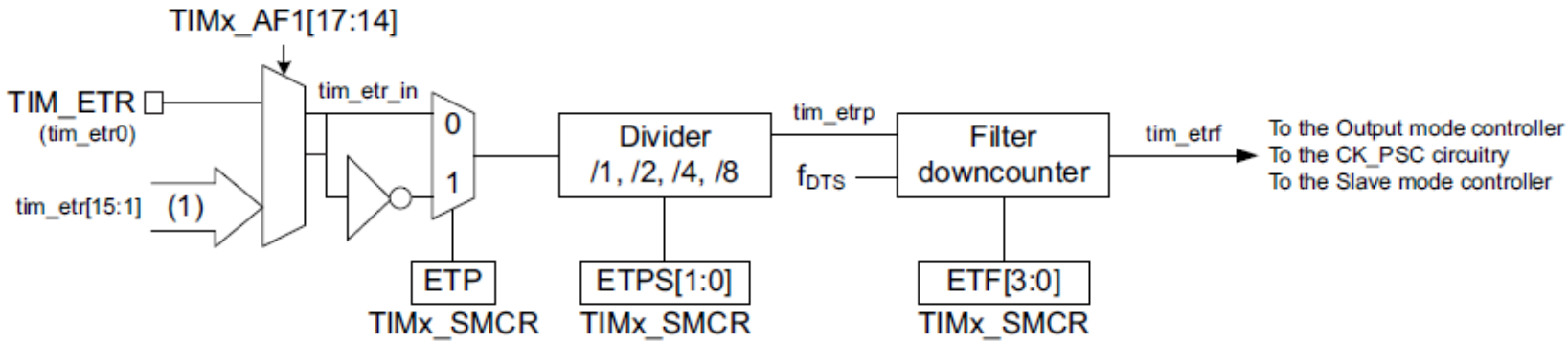
Timers : simple counter

To count external event

Clock source is ETR2

- Can be filtered
- Can be divided
- Can generate interrupt





Timers : simple counter

Start the timer and stop the timer

```
...
/*Start the timer */
HAL_TIM_Base_Start(&htim3);
...
HAL_TIM_Base_Stop(&htim3);
...
```

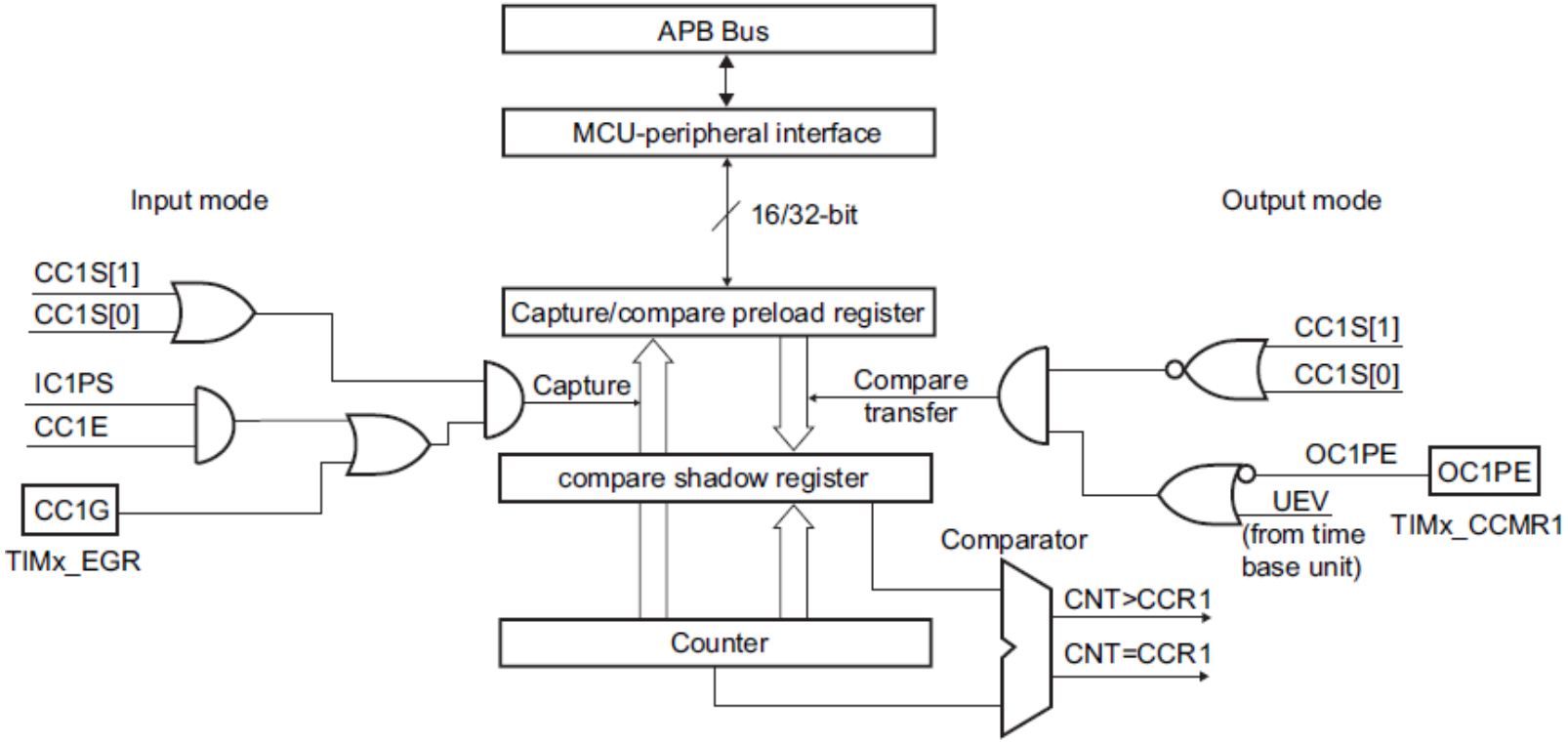
main.c

To read and to write the counter

```
...
  Value = TIM3->CNT;                        // direct access to register
  __HAL_TIM_GET_COUNTER(&htim3);            // alternate access
  TIM3->CNT = 125;                          // direct access to register
   __HAL_TIM_SET_COUNTER(&htim3,125);       // alternate access
...
```

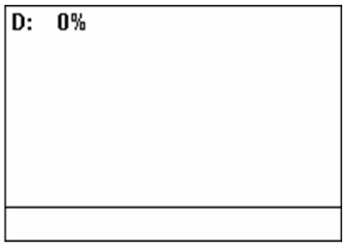
Timers : capture / compare

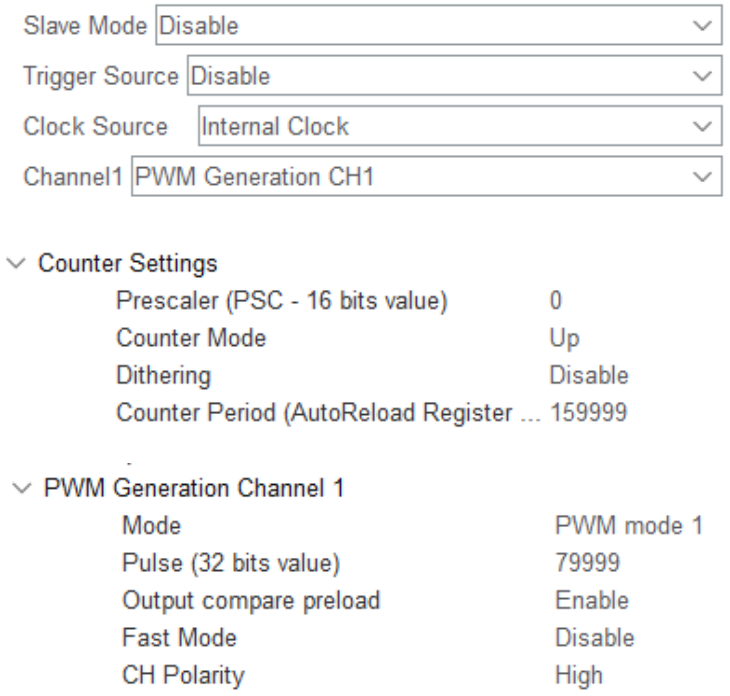
- To capture time from external event
- To compare counter and set an output
  - Clock source can be selected
  - Can generate interrupt



Timers : PWM mode

To generate a PWM





- **Frequency can be selected**
- **Duty can be selected**

What is the frequency and the duty in the example above ?

Timers : PWM mode

Start the timer and stop the PWM

```
...
/*Start and stop the PWM */
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_1);
HAL_TIM_PWM_Stop(&htim3, TIM_CHANNEL_1);
...
```

main.c

To modify the duty cycle

```
...
TIM3->CCR1 = 39999;                 // modify the duty cycle
__HAL_TIM_SET_COMPARE(&htim3,TIM_CHANNEL_1,39999); // alternate
TIM3->ARR = 79999;  // modify the frequency, see the caution below
__HAL_TIM_SET_AUTORELOAD(&htim3, 79999); // alternate
...
```

Caution, if you modify the frequency, the duty cycle is modified

Timer : exercise 1

What is the maximum time between two interrupts of TIM3 when Fosc of the STM32 is 160 MHz?
You know that PSC (prescaler) is a 16 bits register
You know the counter is a 32 bits counter

- **Answer : _____**

If we want an interrupt each 10ms when Fosc is 160MHz, how to configure timer ?

- **Prescaler : _____**
- **Counter period : _____**

Timers : exercise 2

Configure TIM3 to have a PWM of 10kHz with a duty cycle of 25 %. Fosc = 160MHz

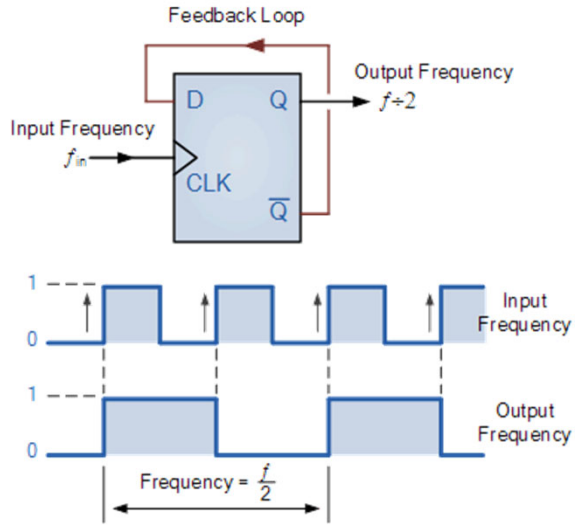
- **Prescaler : _____**
- **Counter period (ARR) : _____**
- **CCR1 : _____**

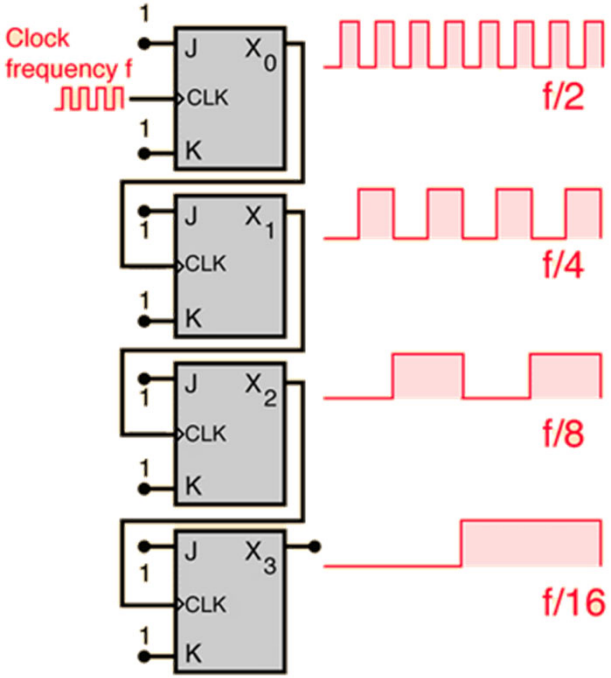
What is the resolution of the PWM?
What is the minimum pulse time (not 0) ?

Timers : the classic crystal 32.768kHz

Why this strange value ?

32768 = 2 x 2 x 2 x 2 x 2 x …… x 2 x 2



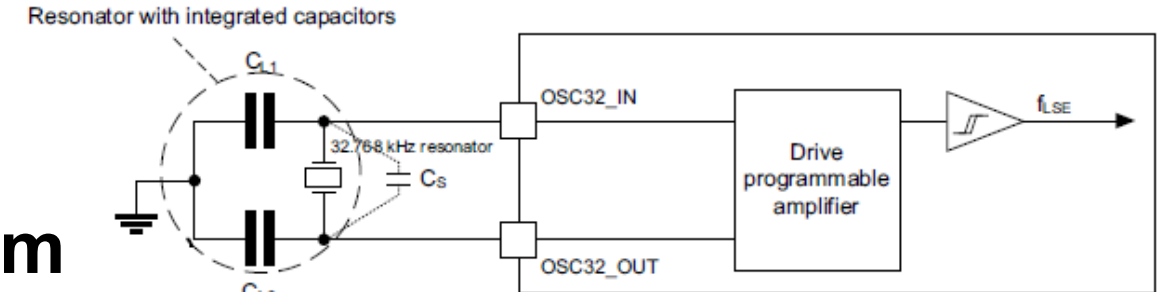


Timers : the classic crystal 32.768kHz

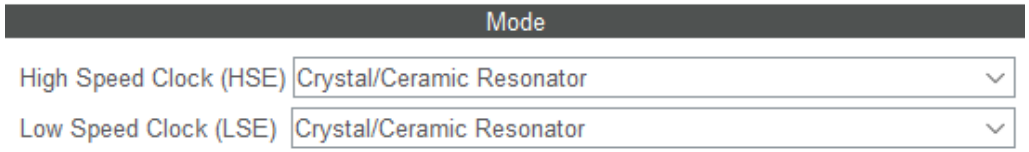
Where and how to choose it?

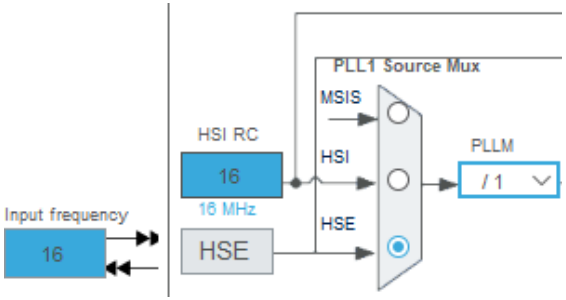
Clock accuracy

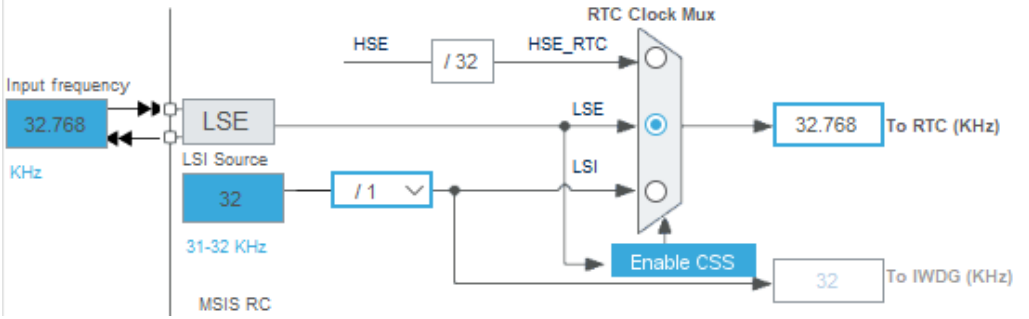
- Crystal 32.768kHz / 5ppm
- Crystal 16MHz / 50 ppm
- STM internal 16MHz (1%)
- Oscillator OCXO 0.05PPB



RCC Mode and Configuration







Timers : the watchdog principle

Watchdog usage

Wake the processor periodically from sleep



Serve as a protection

- If stay in an infinite loop, CPU is reset





Timers : the watchdog usage

Watchdog usage

```
...
/*Reload the watchdog to prevent reset */
HAL_IWDG_RELOAD_COUNTER(hiwdg);

/*to prevent watchdog running in debug mode */
__HAL_DBGMCU_FREEZE_IWDG();

/*the EarlyWakeup callback if used  */
__weak void HAL_IWDG_EarlyWakeupCallback(IWDG_HandleTypeDef *hiwdg)
...
```

Early wakeup can be used of not, if yes, its counter must be smaller than watchdog counter

Timers : the independent watchdog (IWDG)

Watchdog content

- Use special internal clock running at 32kHz on a 12 bits counter
- Can be enabled or disabled (debug mode)
- Divider (4 to 1024)

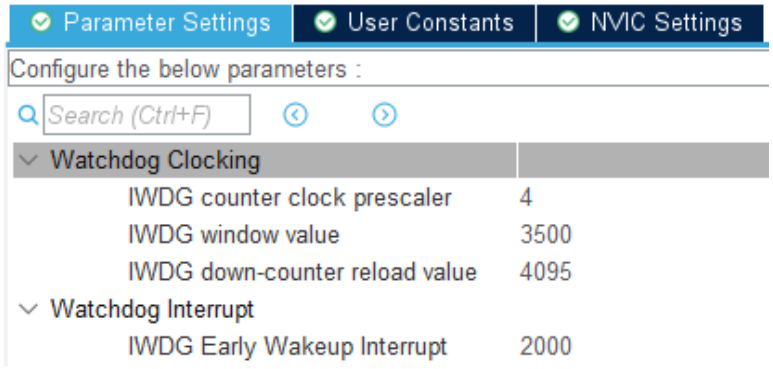
What is the maximum watchdog period ?
Period is : _____

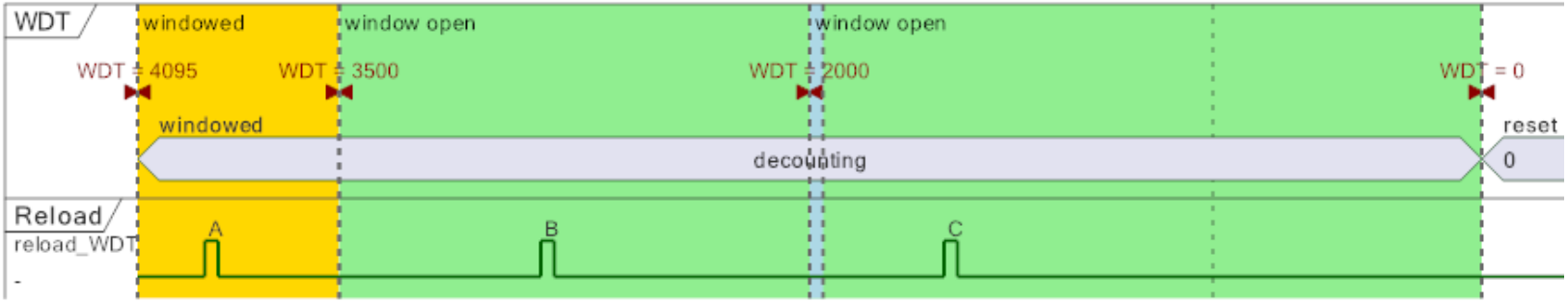
Timers : the independent watchdog (IWDG)

Watchdog details

- **What will arrive on cases A, B, C or none ?**

NB: if a reload is made out of window, a reset is generated.





Time [ms]